MATHIA

MATEMATIČKA ANALIZA 2

Formule — Redovi, integrali, kompleksna analiza, Furije i Laplas

I deo
Brojni redovi

II deo
Funkcionalni i stepeni redovi

III deo
Dvostruki integral

IV deo
Krivolinijski integral

V deo
Kompleksna analiza

VI deo
Furijeovi redovi i Laplasova transformacija

Uči s ljubavlju

Tvoja profesorka

I deo Brojni redovi

Konvergencija i kriterijumi

$$\sum_{n=1}^{\infty} a_n \text{ konvergira } \Rightarrow \lim_{n\to\infty} a_n = 0 \quad \text{(potreban uslov)}$$

$$\text{Dalamber: } \lim \left|\frac{a_{n+1}}{a_n}\right| = L \qquad \text{Ko0̌161i: } \lim \sqrt[n]{|a_n|} = L; \quad L < 1 \Rightarrow \text{konv.}, \ L > 1 \Rightarrow \text{div.}$$

Geometrijski i p-red

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q} \ (|q| < 1) \qquad \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ konvergira } \iff p > 1$$

II deo Funkcionalni i stepeni redovi

Stepeni red i poluprečnik

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n \qquad R = \lim \left|\frac{a_n}{a_{n+1}}\right| \quad \text{ili} \quad \frac{1}{R} = \lim \sqrt[n]{|a_n|}$$

Tejlorov / Maklorenov red

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \qquad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \qquad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

III deo Dvostruki integral

Računanje (Fubini)

$$\iint_D f\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$$

Polarne koordinate

$$x = r\cos\varphi, \quad y = r\sin\varphi \qquad \iint_D f\, dx\, dy = \iint f(r\cos\varphi, r\sin\varphi)\, r\, dr\, d\varphi$$

IV deo Krivolinijski integral

I i II vrste

$$\int_C f\, ds = \int_a^b f(\vec{r}(t))\, |\vec{r}\,'(t)|\, dt \qquad \int_C P\, dx + Q\, dy$$

Grinova formula

$$\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

V deo Kompleksna analiza

Koši–Rimanovi uslovi

$$f = u + iv \text{ analitiǑ10dka } \Rightarrow u_x = v_y, \quad u_y = -v_x$$

Košijeva teorema i reziduumi

$$\oint_C f(z)\,dz = 2\pi i \sum \mathrm{Res} \qquad \mathrm{Res}_{z_0} f = \lim_{z \to z_0} (z - z_0) f(z) \text{ (prost pol)}$$

VI deo Furijeovi redovi i Laplasova transformacija

Furijeov red

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos nx + b_n \sin nx\right), \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f \cos nx\,dx, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f \sin nx\,dx$$

Laplasova transformacija

$$\mathcal{L}\{f\}(s) = \int_0^{\infty} f(t) e^{-st}\,dt \qquad \mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad \mathcal{L}\{f'\} = sF(s) - f(0)$$